Prompt-Based Knowledge Injection for LLM Reranking in Biomedical Information Retrieval

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Abstract

Large Language Models (LLMs) are increasingly used as zero-shot relevance judges in Information Retrieval. In specialized domains such as biomedicine, however, relevance assessment often requires fine-grained conceptual distinctions that general-purpose LLMs may not reliably capture from parametric knowledge alone. In this extended abstract, we summarize our work accepted at CLiC-it 2026 on prompt-based semantic grounding for biomedical document reranking. Rather than relying on domain-specific model adaptation, we dynamically enrich the reranking prompt with curated knowledge from the Medical Subject Headings (MeSH) ontology. For each query, relevant MeSH descriptors are retrieved and their scope notes and narrower terms are injected as supportive context for semantic disambiguation. Experiments on BioASQ show that MeSH-grounded prompting improves over the ungrounded LLM and the considered baselines, with the largest gains on retrieval-difficult queries. An ablation analysis further shows that excessive ontology context can degrade ranking effectiveness. These results suggest that controlled inference-time ontology grounding can provide a lightweight mechanism for incorporating domain knowledge into LLM-based biomedical reranking.

Keywords

Biomedical Information Retrieval, Large Language Models, Knowledge Injection, Re-Ranking, MeSH Ontology

1. Background and Motivation

Biomedical Information Retrieval serves as a foundation for Evidence-Based Medicine tasks, where high-precision ranking is particularly important. The medical lexicon is notoriously challenging, characterized by severe synonymy, ambiguous acronyms, and deep hierarchical specificity that creates a significant vocabulary mismatch [1]. While dense retrieval models are highly efficient, bi-encoder architectures represent each input independently in a fixed-size vector space, which may obscure fine-grained relevance constraints [2]. To mitigate this, modern IR pipelines often employ specialized cross-encoders (e.g., MedCPT [3]) or LLM-based rerankers to perform fine-grained relevance assessment over retrieved candidates.

However, general-purpose LLMs primarily rely on parametric knowledge rather than curated domain definitions and may therefore conflate clinically distinct entities sharing similar lexical surface forms. For example, for the query *"Is mucopolysaccharidosis-plus syndrome (MPSPS) X-linked?"*, an ungrounded LLM may spuriously associate "MPS" with the related *Mucopolysaccharidosis II* (which is X-linked) and incorrectly assign high relevance to an off-target document.

To bridge this knowledge gap without the computational burden of domain-specific fine-tuning, our CLiC-it 2026 work [4] shifts the focus of semantic grounding from static model weights to dynamic query contextualization. Specifically, we designed a **Prompt-Based Knowledge Injection** strategy leveraging the *Medical Subject Headings (MeSH)* ontology. MeSH is a manually curated thesaurus for the life sciences, organizing biomedical concepts within a hierarchical structure and providing explicit textual definitions (*scope notes*) for its descriptors. The proposed reranking strategy is designed for cascaded retrieval over large biomedical collections. In our setting, a hybrid lexical–dense stage first retrieves a high-recall candidate pool from the approximately 38M PubMed records, after which the MeSH-grounded LLM is applied only to the top-ranked candidates [5]. Importantly, the ontology does not introduce additional documents or answer-bearing evidence: it provides query-side semantic

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context, while relevance remains grounded in the candidate abstract. Although such high-precision retrieval can support downstream QA and RAG pipelines [6], the evaluation presented here focuses exclusively on document reranking. This extended abstract summarizes the results of our forthcoming CLiC-it 2026 paper [4] and introduces no additional experimental results.

2. The Knowledge-Grounded Framework

To balance high-recall retrieval with deep semantic reasoning within computational limits, our framework operates in three cascaded stages, as illustrated in Figure 1.

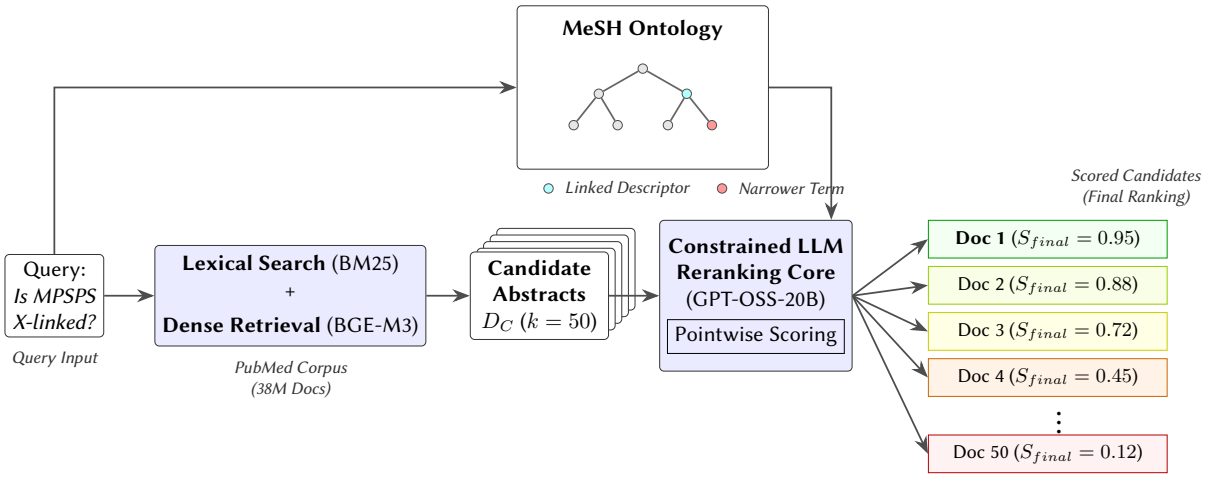


Figure 1: Overview of the proposed knowledge-grounded reranking pipeline. A hybrid retriever first selects candidate PubMed abstracts. The query is then linked to relevant MeSH descriptors, whose scope notes and narrower terms are verbalized and injected into the prompt of an LLM-based relevance judge.

Stage 1: Hybrid Candidate Generation. To address vocabulary mismatch over the full PubMed corpus while bounding inference time [5], we fuse sparse signals (BM25 [7]) with a dense embedding model (BGE-M3 [8]). This step serves as a high-recall filter, passing only the top-k (with $k = 50$) most promising candidates to the reasoning phase.

Stage 2: Dynamic Knowledge Injection. Using the lightweight `all-MiniLM-L6-v2` sentence transformer, we model MeSH concept linking as a semantic similarity task. We retrieve the most relevant MeSH descriptors above a cosine similarity threshold $\tau = 0.45$, together with their definitions (*scope notes*) and directly subordinate taxonomy nodes (*narrower terms*). Both the threshold and the number of injected descriptors are selected on the BioASQ 2024 development set; the final configuration uses $N = 5$. This structural context is linearized into a textual block and injected into the LLM prompt. Crucially, the ontological context is constructed once per query, acting as a semantic cue to support disambiguation rather than as absolute ground truth.

Stage 3: Constrained Generative Re-Ranking. An instruction-tuned open-weight LLM, GPT-OSS-20B [9], acts as a pointwise relevance judge. The prompt instructs the model to use MeSH definitions solely for semantic disambiguation and the interpretation of hierarchical relations, while avoiding factual claims not supported by the candidate abstract. The LLM produces a continuous relevance score, which is linearly interpolated with the upstream hybrid retrieval score to retain the complementary lexical and dense retrieval signals.

Computational Overhead. While prompt-based grounding avoids additional model training, injecting MeSH concepts increases the input context length and consequently the computational cost of LLM

inference. In our pointwise setting, the same query-side MeSH context is provided when scoring each of the 50 candidate abstracts. Since this context is shared across candidates, inference optimizations such as prefix caching could reduce redundant computation. However, we do not report a dedicated latency or throughput analysis, and quantifying the computational trade-off between ranking effectiveness and inference cost remains an important direction for future work.

3. Experimental Evaluation

We evaluated our framework on the BioASQ Task 13b 2025 test set [10], comprising 340 expert-annotated questions over the approximately 38M-record PubMed collection. For all reranking configurations, the common input consists of the top-50 candidates produced by the upstream lexical–dense retrieval stage. For MedCPT and the LLM-based rerankers, we report both **Pure** scores, using the reranker output alone, and **Interpolated** scores ($\alpha = 0.3$), combining the reranker score with the initial Hybrid retrieval signal. The Pure setting corresponds to $\alpha = 1.0$.

Overall Performance. Table 1 shows that, in this candidate-pool setting, the ungrounded **LLM Base** already outperforms the domain-specific MedCPT cross-encoder. More importantly, the proposed **LLM MeSH** configuration provides further gains across all reported cut-offs. The improvement is largest at the top of the ranking, where interpolated MAP@3 increases from 62.14 to 64.69. This suggests that the injected ontology-derived context can help the relevance judge distinguish fine-grained concepts.

Table 1
BioASQ 2025 Test Set Results. **Interp.** denotes the interpolated score ($\alpha = 0.3$).

	Baselines				Ours
Metric (Mode)	**BM25**	**Hybrid**	**MedCPT**	**Base**	**MeSH**
MAP@10 (Pure)	34.79	–	38.46	48.25	49.14
MAP@10 (Interp.)	34.79	44.82	45.27	50.67	**52.26**
MAP@5 (Pure)	39.25	–	41.61	54.51	56.66
MAP@5 (Interp.)	39.25	51.03	51.64	57.84	**59.75**
MAP@3 (Pure)	42.61	–	44.68	59.67	61.57
MAP@3 (Interp.)	42.61	56.01	56.76	62.14	**64.69**

Impact by Query Difficulty. As shown in Table 2, the benefit of MeSH injection increases as upstream retrieval becomes more difficult. On *Hard* queries, grounding improves MAP@3 by +4.52, whereas the gain decreases to +0.88 on *Simple* queries, where the ungrounded model already performs strongly.

Table 2
Performance (MAP@3) stratified by query difficulty. The analysis includes the 328 queries with $R@50 > 0$; the 12 queries with $R@50 = 0$ are excluded because their candidate pools contain no relevant documents that can be recovered by reranking.

Difficulty (Recall@50)	**Count**	**Base LLM**	**MeSH LLM**	**Δ Gain**
Hard (0%–20%]	59	40.21	**44.73**	+4.52
Medium (20%–70%]	167	58.05	**61.11**	+3.06
Simple (70%–100%]	102	88.83	**89.71**	+0.88

Ablation on Context Saturation. A sensitivity analysis on the BioASQ development set shows a non-monotonic relationship between the amount of injected knowledge and ranking effectiveness. MAP@3 increases from 63.81 at $N = 3$ to 66.71 at $N = 5$, where the inclusion of narrower terms provides the best overall configuration. Increasing the context to $N = 10$ instead reduces MAP@3 to 61.83. At this larger context size, peripheral descriptors and hierarchical terms may dilute the semantic focus of the prompt and induce semantic drift. These results indicate that ontology grounding benefits from selective context construction rather than unrestricted knowledge injection.

4. Conclusion

We presented a prompt-based MeSH injection strategy for biomedical document reranking, summarizing our recent work on inference-time ontology grounding [4]. In the BioASQ setting, MeSH-grounded prompting improves over the ungrounded LLM across the reported ranking cut-offs, with the largest gains on retrieval-difficult queries. At the same time, the ablation analysis shows that increasing the amount of injected knowledge is not monotonically beneficial: excessive ontological context can introduce semantic noise and degrade ranking quality. These findings support inference-time ontology grounding as a lightweight mechanism for incorporating explicit domain knowledge into LLM relevance judgments without additional model training. The present evidence is limited to BioASQ, one open-weight LLM, and one lightweight concept-linking strategy; evaluating different model families, concept linkers, biomedical benchmarks, and ontologies represents the natural next step.

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Declaration on Generative AI

During the preparation of this manuscript, the authors used OpenAI's ChatGPT for English proofreading and minor stylistic improvements of selected sentences. All AI-assisted edits were reviewed and validated by the authors, who take full responsibility for the final content.

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